SLO Strategy Analysis

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Natalya Langford

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1. Summary

This report analyzes historical option trades across both call and put strategies to evaluate how strike selection (T0 POTM) impacts profitability, consistency, and risk.

The data shows that:

- A majority of trades are placed within lower POTM ranges (0–10%)
- These trades produce disproportionately high average losses and extreme tail-risk events
- Increasing POTM distance significantly improves win rate and reduces both average and catastrophic losses

The results suggest that the current strategy prioritizes premium income at the cost of elevated risk. A modest shift toward higher POTM ranges (15–20%) has the potential to substantially improve consistency and reduce downside exposure across both strategies.

2. Terminology

SLO: Sophia Langford Options
POTM: Percent out of money
APR: Annual percentage rate
T0: Time zero - what the metric was at opening

3. Methodology

The dataset size is sufficient to capture both typical outcomes and rare tail-risk events, allowing for meaningful statistical analysis of strategy performance.
Data includes all historical option trades.

- Trades were grouped by **T0 POTM (distance from strike)**
- Metrics analyzed:
    - Assignment loss
    - Total loss (including buy-to-close)
    - Combined loss
    - Win rate
    - Close APR
- Results are aggregated by POTM bucket.

Particular attention is given to tail-risk events (extreme losses), as these outcomes disproportionately impact long-term strategy performance.

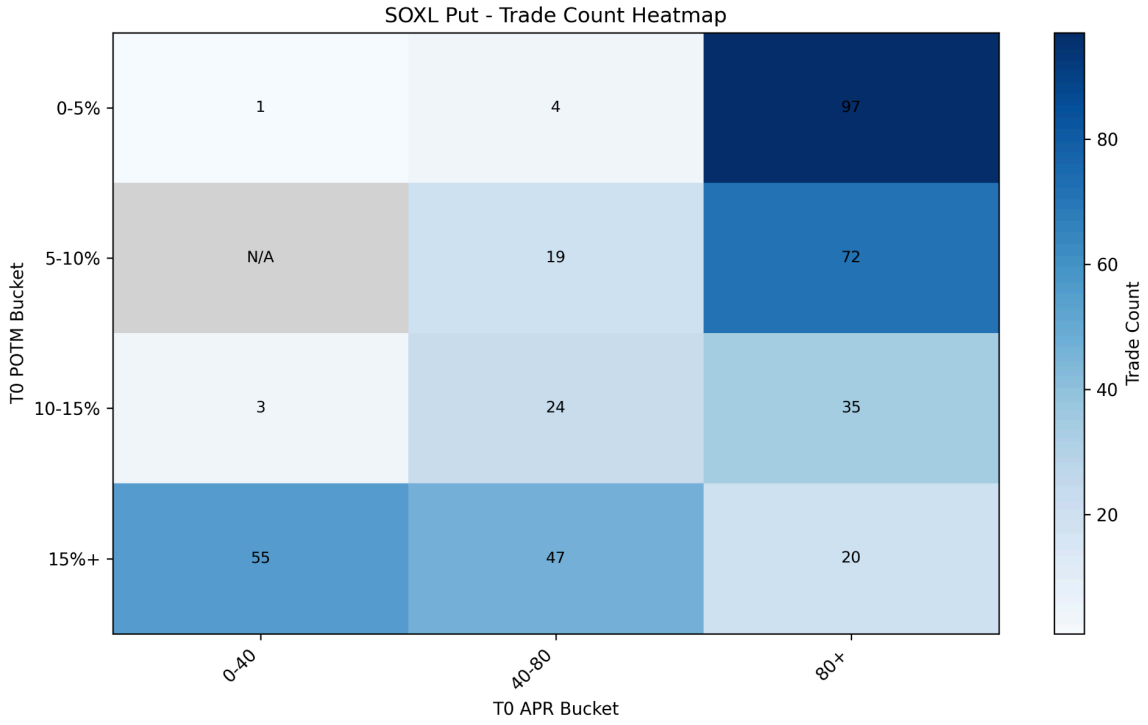
4. Put Strategy Analysis

4.1. Put Strategy Overview

The put-selling strategy analyzed in this report focuses on selling cash-secured puts across a range of strike distances (T0 POTM) and premium levels (APR). The goal of this strategy is to generate consistent income while minimizing assignment risk and drawdowns.

It is important to note that this analysis focuses on ETF-based trades, specifically SOXL and TQQQ, which are leveraged instruments and exhibit higher volatility than typical equities. While these characteristics may amplify both returns and risk, the structural patterns observed in this analysis—particularly the relationship between POTM selection, win rate, and performance—are expected to generalize across similar option-selling strategies.

4.2. Trade Distribution by POTM and APR

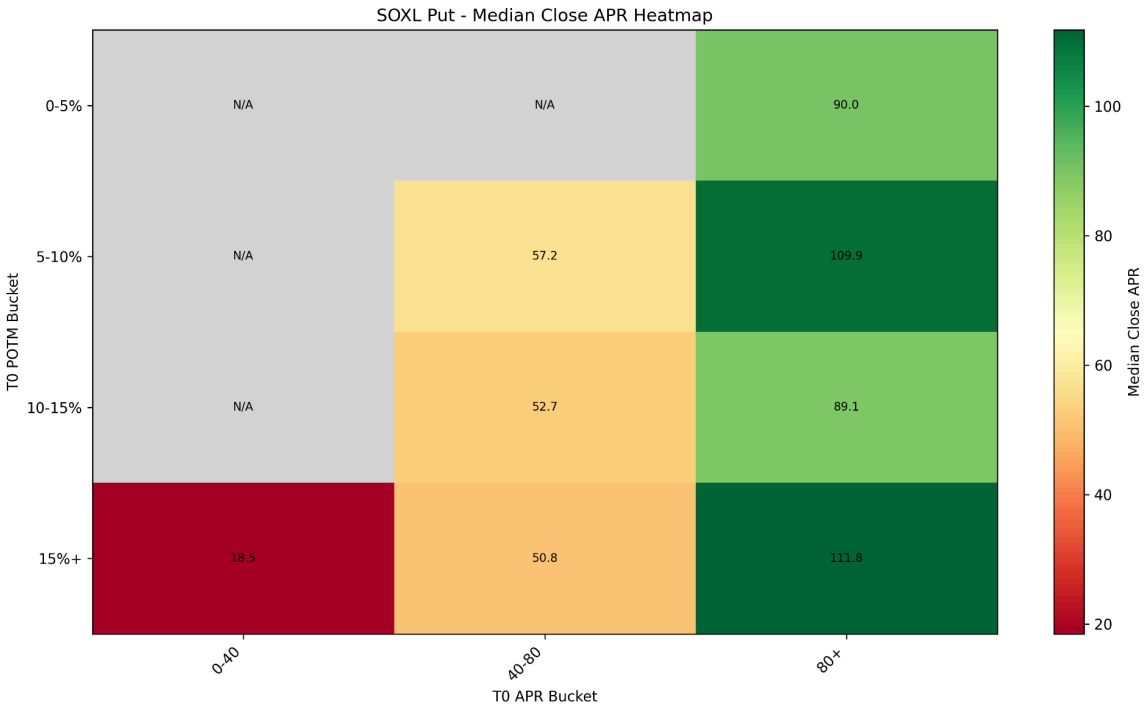


- The trade count heatmap shows how trades are distributed across different levels of POTM (distance from the strike price) and APR (premium yield).

  The majority of trades are concentrated in the **0–5% POTM and 80%+ APR bucket**, indicating a strong preference for high-premium, near-the-money positions. Moderate activity is also observed in the **5–15% POTM range**, particularly in the 40–80% APR category. In contrast, deeper out-of-the-money positions (15%+) are less frequently traded at high APR levels.

  This distribution suggests a tendency to prioritize premium generation over distance from the strike, resulting in a higher concentration of trades in riskier positioning zones. This trade-off increases exposure to adverse price movement despite higher initial income.

4.3. Profitability (Median Close APR)

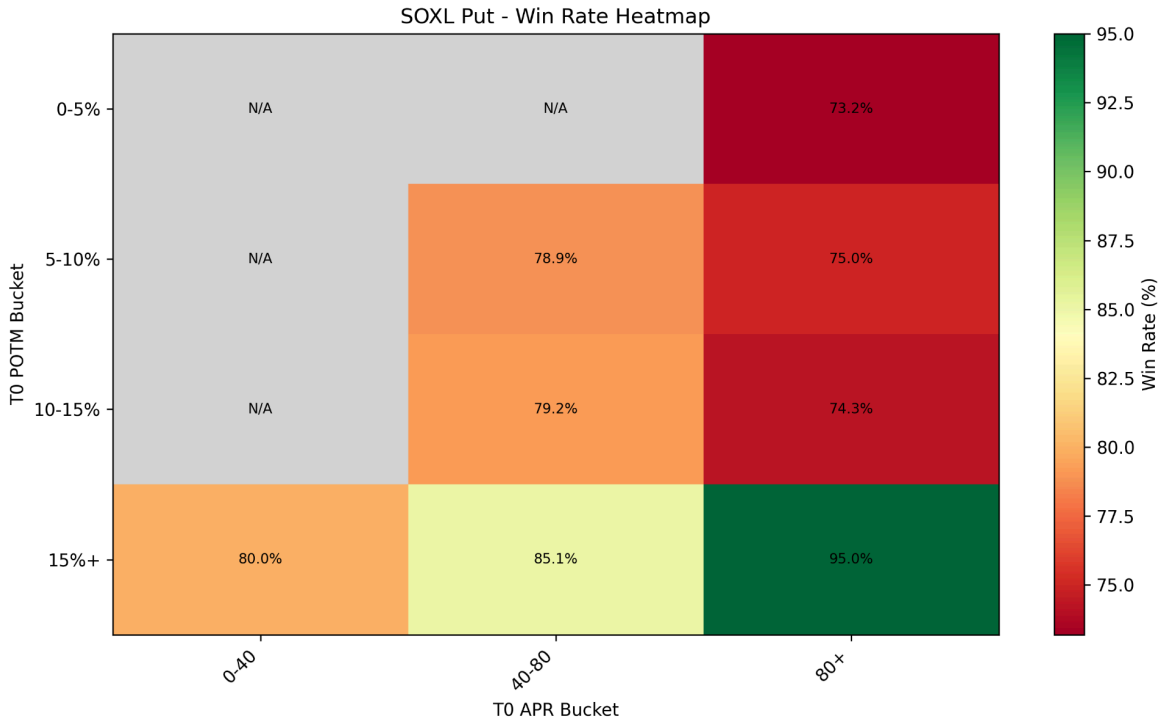


- The median close APR heatmap evaluates the profitability of trades across POTM and APR buckets.

  Higher APR trades (80%+) consistently deliver strong returns across most POTM levels, with median APR values reaching approximately **90–112%**. Moderate APR trades (40–80%) produce stable but lower returns in the **50–57% range**, while low APR trades (0–40%) show limited profitability and sparse participation.

Notably, profitability remains relatively strong even at higher POTM levels, indicating that moving further out-of-the-money does not necessarily sacrifice return potential when paired with appropriate premium levels.

4.4. Win Rate Analysis



- The win rate heatmap highlights the consistency of outcomes across different trade configurations.

  A clear trend emerges where **win rate increases as POTM increases**. Deeper out-of-the-money trades (15%+) achieve the highest win rates, reaching approximately **85–95%**, particularly in higher APR environments. In contrast, trades closer to the money (0–10% POTM) exhibit lower win rates, generally ranging between **73–79%**.

  This demonstrates that increasing distance from the strike significantly improves trade consistency and reduces the likelihood of unfavorable outcomes.

4.5. Key Insights

The analysis reveals several important patterns:

- Higher POTM improves consistency: Trades placed further out-of-the-money consistently achieve higher win rates.
- High APR does not necessarily imply poor performance: Even high-premium trades can maintain strong outcomes when combined with appropriate strike selection.
- Risk concentration exists in low POTM trades: The most frequently traded positions (0–5% POTM) are also associated with lower win rates and higher exposure to adverse price movement.
- Balanced positioning is more effective: Combining moderate-to-high POTM (10–15%+) with strong premium levels provides a more favorable balance between return and consistency.

The put-selling strategy demonstrates strong income-generating potential, particularly when trades are structured with sufficient distance from the strike price. While high-premium, near-the-money trades offer attractive returns, they come with reduced consistency and increased risk.

In contrast, deeper out-of-the-money trades provide significantly higher win rates while still maintaining competitive returns, especially in higher APR environments. These findings strongly suggest that consistent profitability is not driven by aggressive premium collection, but by disciplined strike selection and risk control.

5. Call Strategy Analysis

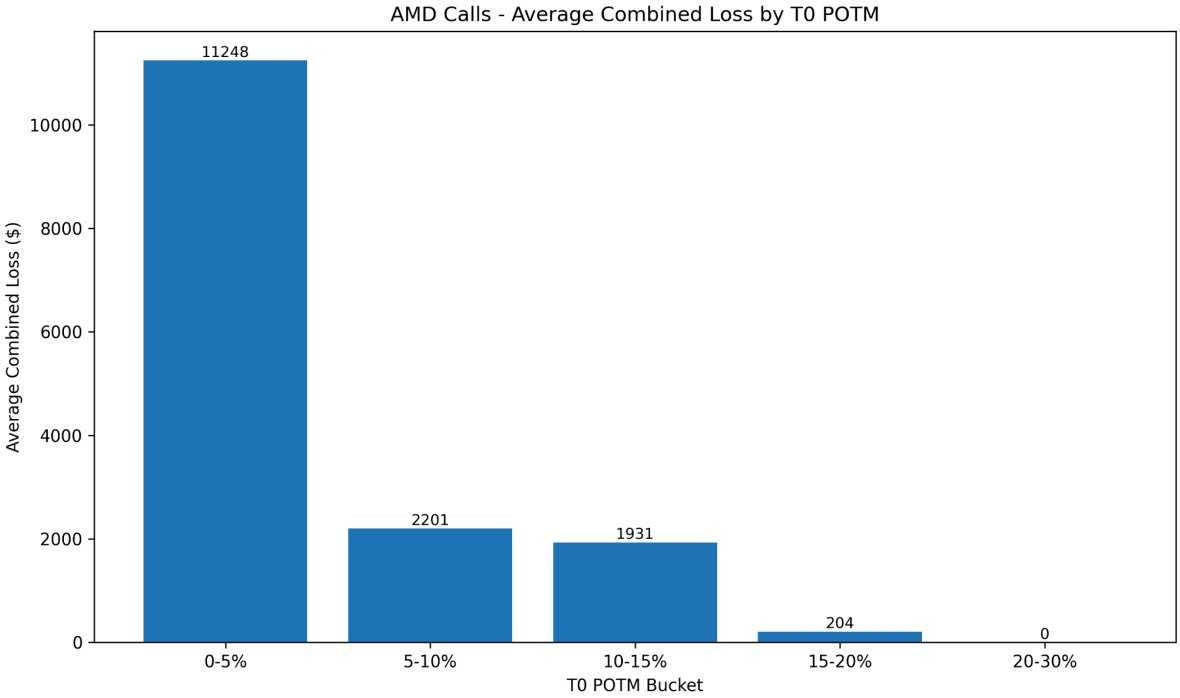
5.1. Trade Distribution (AMD)

POTM Range	Trade Count	Percentage
0-5%	173	35.9%
5-10%	147	30.5%
10-15%	98	20.3%
15-20%	41	8.5%
20-30%	13	2.7%
30%+	10	2.1%

- The majority of AMD call trades (approximately 66%) are executed within 0–10% out-of-the-money (POTM), with over 85% placed within 15% POTM. This indicates a strong concentration in lower-distance strike selections.
- This distribution shows that most trades are concentrated in the 0–10% POTM range, which aligns with the highest-risk region identified in the loss analysis. In contrast, the 15–20% POTM range—associated with significantly lower average and maximum losses—is used in less than 10% of trades.
- This imbalance suggests that the strategy is systematically concentrated in the highest-risk region identified by the data.

The following analysis examines how loss behavior changes across different POTM ranges.

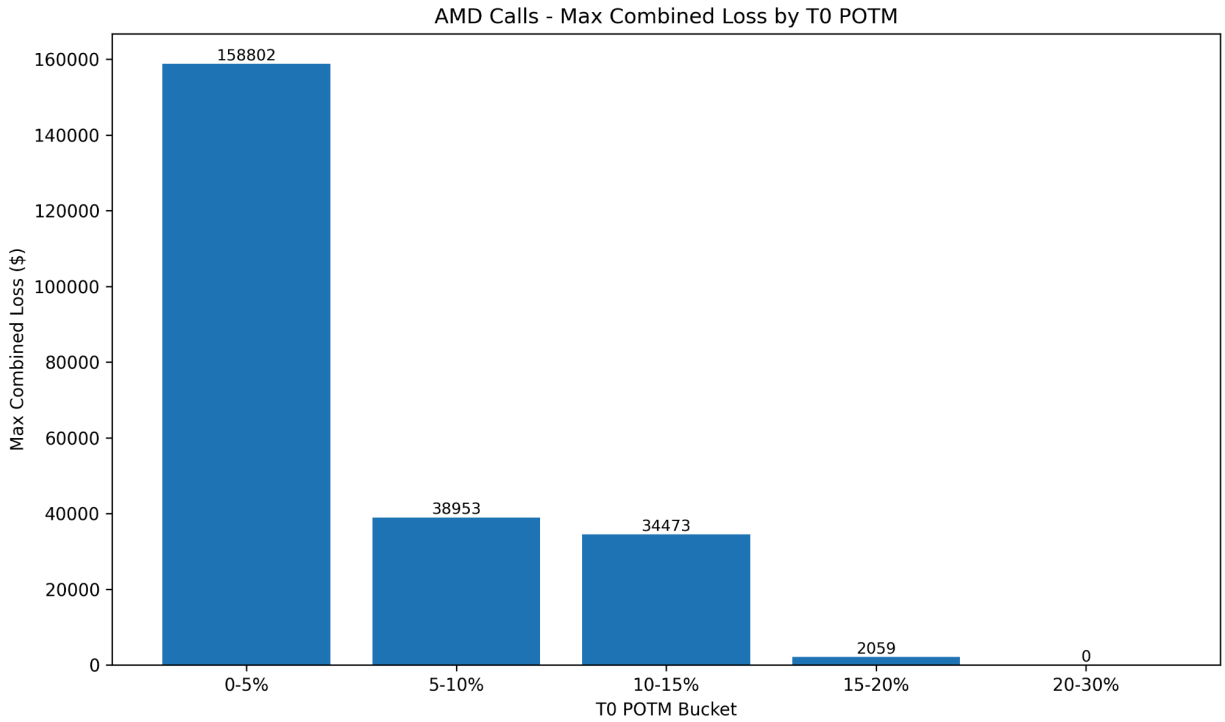
5.2. AMD Calls - Average Combined Loss



- This chart shows the *typical loss per trade* for AMD call positions at different levels of T0 POTM. The results clearly indicate that lower POTM trades, especially in the 0–5% range—are associated with significantly higher average losses, exceeding $11,000 per trade. In contrast, trades placed in the 15–20% POTM range exhibit dramatically lower average losses, closer to a few hundred dollars. This suggests that while lower POTM trades may appear attractive due to higher premium collection, they expose the strategy to substantially greater downside risk.

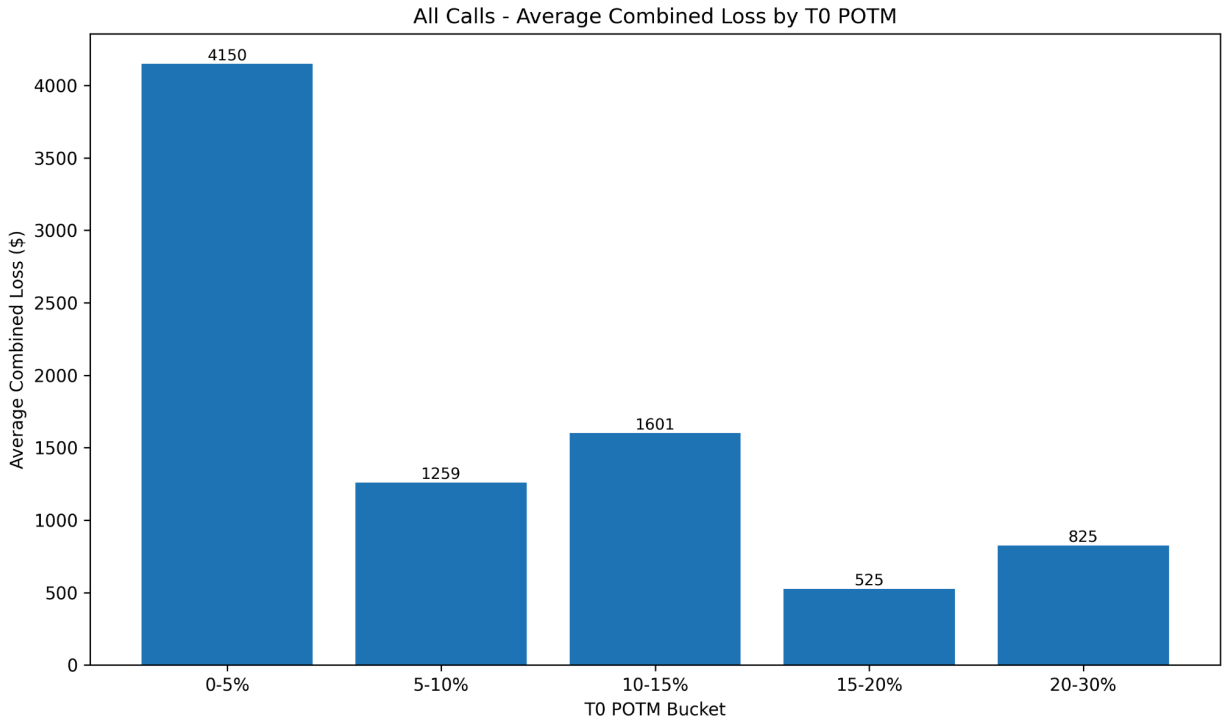
Moving slightly further out-of-the-money materially reduces average losses while still allowing for consistent premium capture. This indicates that strike selection, not market direction, is the primary driver of risk in this strategy.

5.3. AMD Calls - Maximum Loss



- While average loss provides insight into typical outcomes, this chart highlights *worst-case risk exposure*. The data shows that the largest losses occur overwhelmingly in lower POTM ranges, with extreme events exceeding $150,000 in the 0–5% bucket. Even in moderately low POTM ranges such as 5–10%, maximum losses remain significantly elevated. In contrast, higher POTM trades (15–20%) demonstrate substantially reduced worst-case losses. This reinforces that the current strategy carries *tail risk*—rare but extremely damaging outcomes—that is heavily concentrated in low POTM positions. From a risk management perspective, reducing exposure to these zones can meaningfully limit catastrophic loss events.

5.4. All Calls - Average Combined Loss by T0 POTM



- To determine whether this pattern is specific to AMD or representative of a broader trend, the same analysis was performed across all call trades. The results show a consistent pattern: lower POTM trades systematically produce higher average losses, while higher POTM trades demonstrate improved loss profiles. This consistency across multiple stocks suggests that the observed risk is not due to isolated cases but is instead inherent to the strategy itself. Therefore, adjusting POTM selection is likely to improve performance not only for AMD trades but across the entire portfolio.

5.5. Key Insights

The analysis consistently shows that lower POTM trades (0–10%) are associated with both higher average losses and significantly larger worst-case outcomes. In contrast, trades in the 15–20% POTM range demonstrate substantially improved risk characteristics. Despite this, the majority of trades remain concentrated in the higher-risk zones. This suggests that adjusting strike selection alone could materially improve overall performance without requiring changes to the core strategy.

6. Cross Strategy Insights

When comparing call and put strategies, a consistent pattern emerges:

- In both strategies, lower POTM positions generate higher risk exposure
- Higher POTM improves consistency and reduces extreme outcomes
- Premium-focused positioning increases vulnerability to large losses

This consistency across fundamentally different strategies indicates that the observed behavior is not market-specific, but structural to the strategy itself.

7. Conclusion & Strategy Recommendation

The analysis across both call and put strategies reveals a consistent and structural pattern:
risk is heavily concentrated in low POTM positions, while higher POTM positions provide significantly improved consistency and reduced exposure to extreme losses.

Despite this, the majority of trades remain concentrated in lower POTM ranges, where both average losses and tail-risk events are most severe.

This indicates that current performance is not limited by market conditions, but by strike selection behavior.

Recommended Adjustment:

- Gradually shift trade placement toward **10–20% POTM**
- Maintain strong premium levels (APR), but avoid prioritizing yield at the expense of positioning
- Reduce exposure to 0–5% POTM trades, where risk is disproportionately high

Expected Outcome:

- Lower average loss per trade
- Significant reduction in catastrophic loss events
- Improved consistency and long-term stability

This adjustment does not require changing the overall strategy, but rather refining execution to better align risk and return.

8. Future Work & Expanded Analysis Plan

While the current analysis provides strong evidence that strike selection (T0 POTM) is the primary driver of risk and performance, it is based on a limited set of historically executed trades.

To further validate and strengthen these findings, the next phase of analysis will involve generating and evaluating synthetic option trades across the full range of possible positions.

8.1 Synthetic Trade Generation

Rather than relying only on trades that were actually executed, synthetic trades will be created to represent all realistic options available on a given trading day.

This includes:

- All available put option strikes and expirations
- Across selected ETFs (initially SOXL and TQQQ)
- Using consistent entry assumptions (e.g., mid-price)

Each synthetic trade will be recorded in a structured format and treated identically to real trades in subsequent analysis.

8.2 Large-Scale Dataset Construction

By generating synthetic trades across multiple days and market conditions, it will be possible to construct a significantly larger dataset representing the full decision space.

This enables:

- Evaluation of strategies that were not previously executed
- Elimination of selection bias in historical trades
- More robust statistical conclusions

8.3 Outcome Evaluation

Each synthetic trade will be evaluated based on its eventual outcome (e.g., expiration value, assignment, return metrics).

The same metrics used in this report will be applied:

- T0 POTM
- T0 APR
- Win rate
- Realized return
- Tail risk exposure

8.4 Strategy Optimization

With a sufficiently large dataset, it will be possible to identify optimal strategy regions across multiple dimensions, including:

- POTM (distance from strike)
- APR (premium yield)
- Time to expiration

This will allow for data-driven recommendations such as:

- Optimal strike selection ranges
- Trade-offs between yield and risk
- Conditions under which higher premium strategies are justified

8.5 Expected Outcome

This expanded analysis will allow conclusions to be drawn with a significantly higher degree of confidence.

Rather than relying solely on past behavior, the strategy can be evaluated across all possible choices, enabling a clear identification of the most consistent and risk-efficient approach. This approach transforms the analysis from observational to exhaustive, allowing strategy decisions to be based on the full range of available opportunities rather than a limited subset of executed trades.